

Set A

$$01. F(A, B, C) = C' + B'C + ABC'$$

$$= (A+A')(B+B')C' + (A+A')B'C + ABC'$$

$$= \underline{ABC'} + AB'C' + A'BC' + A'B'C' + AB'C + A'B'C + \underline{ABC'}$$

$$= ABC' + AB'C' + A'BC' + A'B'C' + AB'C + A'B'C$$

$$= 110, 100, 010, 000, 101, 001$$

$$= \sum (0, 1, 2, 4, 5, 6)$$

$$02. F(X, Y, Z, W) = (X+Z'W)(X'+Y')$$

$$= (X+Z')(X+W)(X'+Y')$$

$$= (X+YY'+Z'+WW')(X+YY'+ZZ'+W)(X'+Y'+ZZ'+WW')$$

$$= \underline{(X+Y+Z'+W)} (X+Y+Z'+W') \underline{(X+Y'+Z'+W)} (X+Y'+Z'+W')$$

$$(X+Y+Z+W) \underline{(X+Y+Z'+W)} (X+Y'+Z+W) \underline{(X+Y'+Z'+W)}$$

$$(X'+Y'+Z+W) (X'+Y'+Z+W') (X'+Y'+Z'+W) (X'+Y'+Z'+W')$$

$$= (X+Y+Z'+W) (X+Y+Z'+W') (X+Y'+Z'+W) (X+Y'+Z'+W') \\ (X+Y+Z+W) (X+Y'+Z+W) (X'+Y'+Z+W) (X'+Y'+Z+W') \\ (X'+Y'+Z'+W) (X'+Y'+Z'+W')$$

$$= 0010, 0011, 0110, 0111, 0000, 0100, 1100,$$

$$1101, 1110, 1111$$

$$= \prod (2, 3, 6, 7, 0, 4, 12, 13, 14, 15)$$

Set B

01. $F(D, A, B, C) = (A' + CD')(A' + B')$

$$= (A' + C)(A' + D')(A' + B')$$

$$= (DD' + A' + BB' + C)(D' + A' + BB' + CC')(DD' + A' + B' + CC')$$

$$= (D + A' + B + C) \underline{(D + A' + B' + C)} \underline{(D' + A' + B + C)} \underline{(D' + A' + B' + C)}$$

$$\underline{(D' + A' + B + C)} \underline{(D' + A' + B + C')} \underline{(D' + A' + B' + C)} \underline{(D' + A' + B' + C')}$$

$$\underline{(D + A' + B' + C)} \underline{(D + A' + B' + C')} \underline{(D' + A' + B' + C)} \underline{(D' + A' + B' + C')}$$

$$= (D + A' + B + C)(D + A' + B' + C)(D' + A' + B + C)(D' + A' + B' + C)$$

$$(D' + A' + B + C')(D' + A' + B' + C')(D + A' + B' + C')$$

$$= 0100, 0110, 1100, 1110, 1101, 1111, 0111$$

$$= \pi(4, 6, 12, 14, 13, 15, 7)$$

02. $F(X, Y, Z) = Z + X'Y + X'YZ$

$$= (X + X')(Y + Y')Z + X'Y(Z + Z') + X'YZ$$

$$= XYZ + XY'Z + \underline{X'YZ} + X'Y'Z + \underline{X'YZ} + X'YZ' + \underline{X'YZ}$$

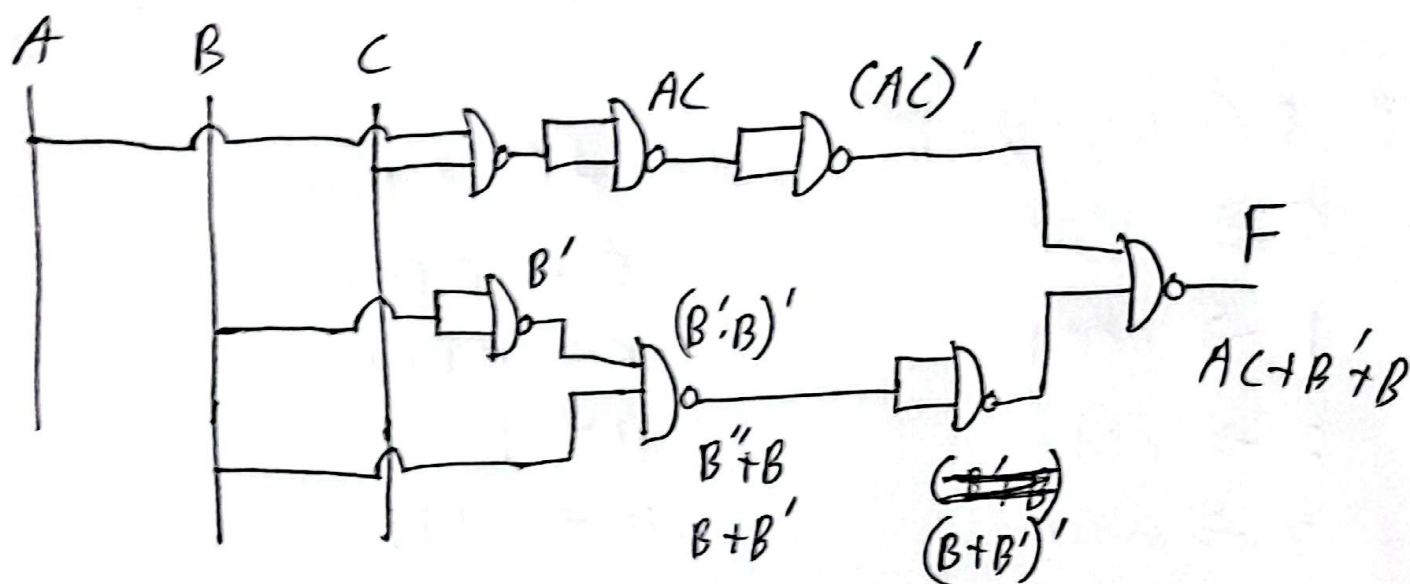
$$= XYZ + XY'Z + \underline{X'YZ} + X'Y'Z + X'YZ'$$

$$= 111, 101, 011, 001, 010$$

$$= \sum(7, 5, 3, 1, 2)$$

Set-A

$$F = AC + B' + B$$



Set-B

$$F = (Y' + X'Z)'$$

